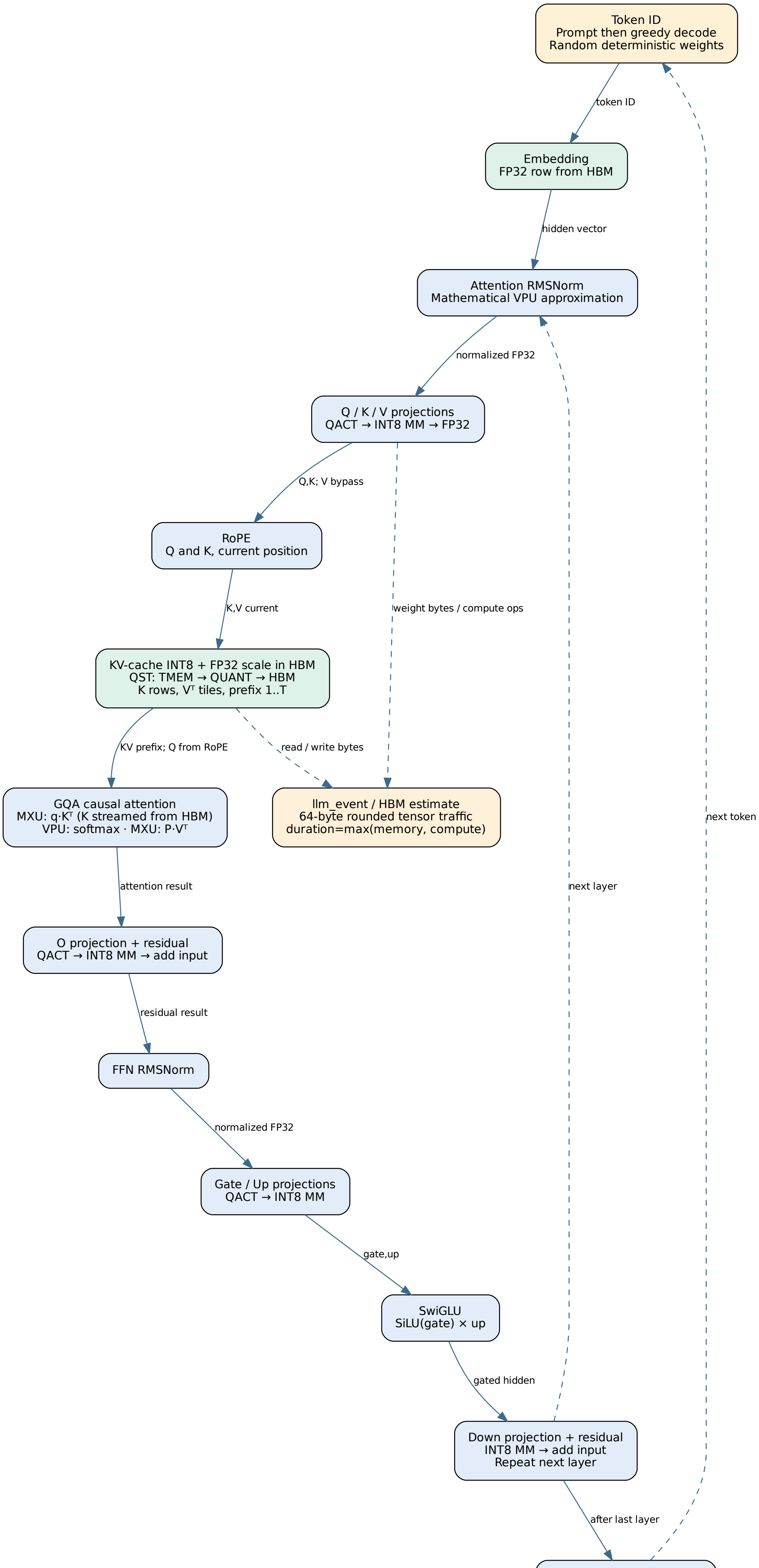03 LLM numerical model